Food Ordering Behavior Analysis

Data Preprocessing, Business Questions & Visualization Report

1. Dataset Overview

The dataset contains 50,000 food ordering records capturing customer demographics, city, cuisine, meal type, restaurant type, ratings, delivery time and order values. The data was analyzed in Tableau to identify customer preferences and business insights.

Column Name	Data Type	Description
Order_ID	Numeric (ID)	Unique identifier for each food order
Customer_Age	Numeric	Age of the customer
Age_Group	Categorical	Gen Z, Millennials, Adults, Child
City	Categorical	City where the order was placed
Cuisine	Categorical	Cuisine ordered
Meal_Type	Categorical	Breakfast, Lunch, Dinner or Snacks
Restaurant_Type	Categorical	Budget, Mid-range or Premium
Company_Type	Categorical	Alone, Friends, Family or Partner
Delivery_Time	Numeric	Delivery time (minutes)
Delivery_Fee	Numeric	Delivery charge
Order_Value	Numeric	Total order value
Customer_Rating	Numeric	Customer rating
Payment_Method	Categorical	Mode of payment

2. Data Preprocessing Steps

Before developing the Tableau dashboard, the dataset was preprocessed to ensure data quality and accuracy. The following steps were performed:

1. Structure Check

Verified that all records and columns were imported correctly with the appropriate data structure.

2. Missing Value Check

Checked the dataset for missing or null values. No significant missing values were identified.

3. Duplicate Record Check

Verified the dataset for duplicate **Order_ID** records and ensured that each order was unique.

4. Data Type Validation

Validated the data types of all variables to ensure accurate analysis.

- **Numeric Fields:** Order Value, Delivery Fee, Delivery Time, Customer Age, Customer Rating
- **Categorical Fields:** City, Cuisine, Meal Type, Restaurant Type, Company Type, Age Group, Payment Method

5. Data Consistency Check

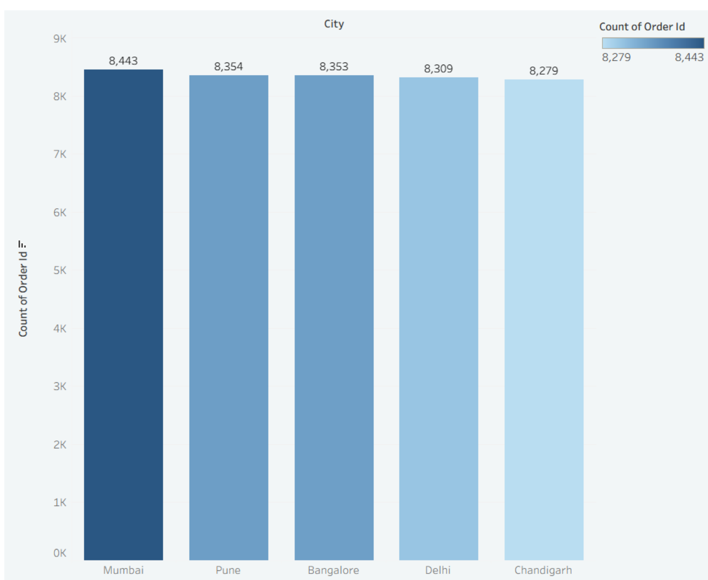
Reviewed categorical fields to ensure consistent naming and formatting across the dataset.

6. Dashboard Readiness

After preprocessing, the dataset was confirmed to be clean, consistent, and ready for visualization in Tableau.

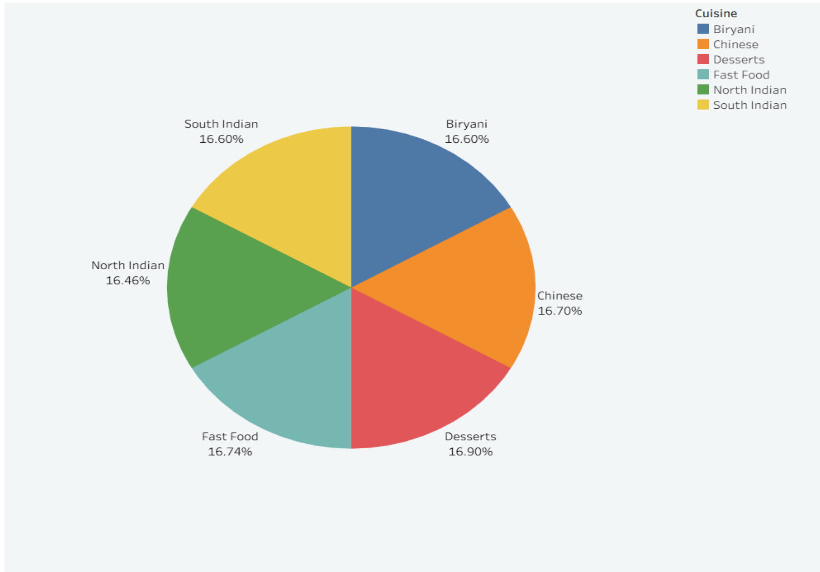
3. Business Questions & Visualizations

Q1. Which city records the highest number of food orders?



The dashboard indicates that **Mumbai** has the highest order volume with **8,443 orders**, followed closely by **Pune (8,354)** and **Bangalore (8,353)**. Chandigarh records the lowest order volume among the top five cities. This suggests that metropolitan cities contribute significantly to customer demand and overall business growth.

Q2. Which cuisine is most preferred by customers?



Customer preferences are distributed fairly evenly across all cuisines. **Desserts (16.90%)** represent the highest preference, closely followed by **Fast Food (16.74%)**, **Chinese (16.70%)**, and **Biryani (16.60%)**. The balanced distribution indicates that customers enjoy a diverse range of cuisines rather than concentrating on a single category.

Q3. Which age group places the highest number of orders?

The dashboard shows that **Millennials** and **Gen Z** contribute the highest number of food orders across all restaurant types. Adults contribute a moderate number of orders, while the Child age group records the lowest order count. This indicates that younger customers form the primary customer base for online food delivery platforms.

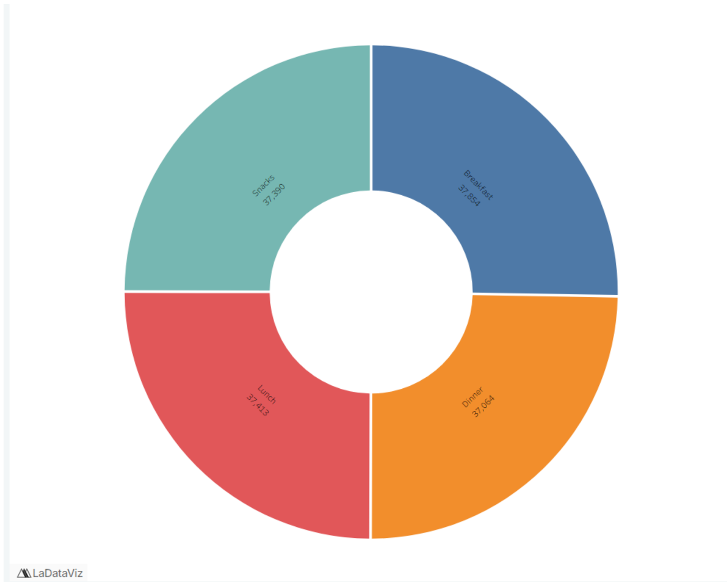
Q4. How does company type influence food ordering?

Customers ordering **alone** place the highest number of food orders (**48,617**), followed by customers ordering with **friends** and **partners**. Orders placed by **families** are comparatively lower. This suggests that individual ordering remains the most common customer behavior in the dataset.

Q5. Which meal type generates the highest revenue?

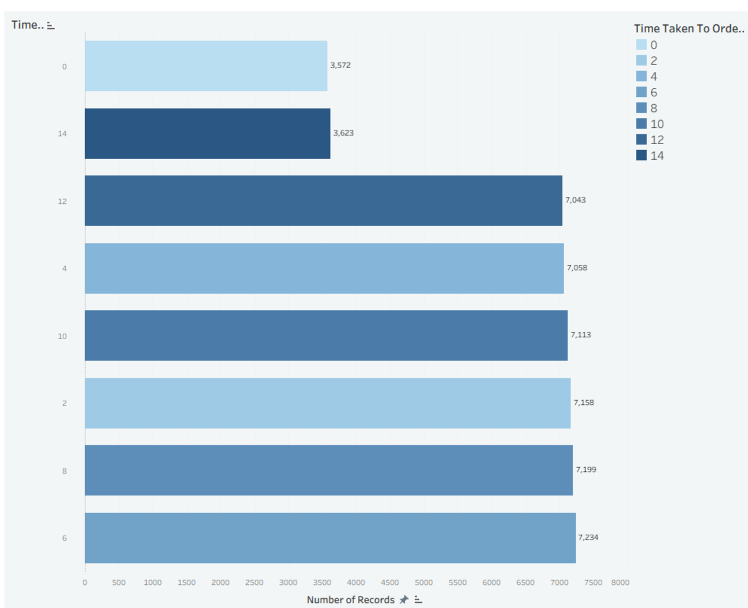
Among all meal categories, **Breakfast** generates the highest revenue (**₹11.83 lakh**), followed by **Dinner**, **Lunch**, and **Snacks**. Although the difference is relatively small, breakfast contributes the greatest share of order value, making it an important revenue-generating meal period.

Q6. How do customer ratings vary across meal types?



Customer ratings remain fairly consistent across Breakfast, Lunch, Dinner and Snacks, indicating that customer satisfaction is maintained throughout different meal periods. No significant variation is observed between meal types.

Q7. How is delivery time distributed?



Most customer orders are delivered within a moderate delivery time range. Delivery durations around **6–10 minutes** account for the largest number of orders, while extremely short and longer delivery times occur less frequently. This indicates stable delivery performance across the dataset.

4. Key Takeaways

- Mumbai has the highest order volume.
- Millennials and Gen Z dominate food ordering.
- Cuisine demand is balanced across categories.
- Breakfast contributes the highest revenue.
- Dashboard supports data-driven business decisions.